LECTURE I

OPS Lecture 1: Introduction to Circuits

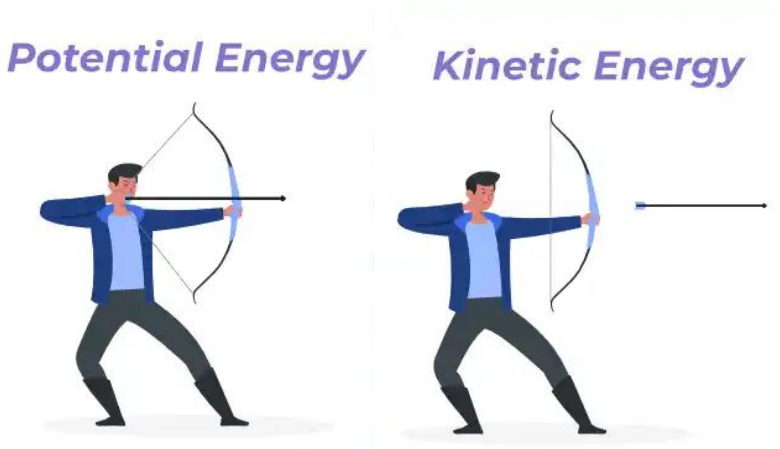
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SECTION I

Ohm's Law

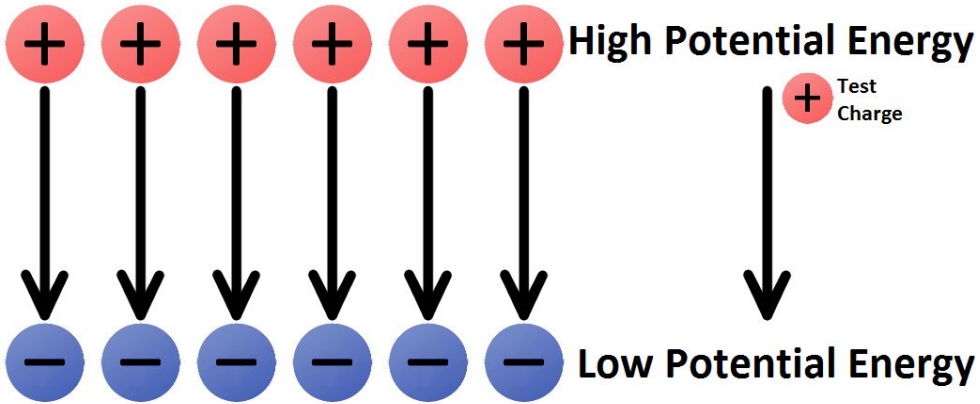
Energy

- **Energy** is the ability to do work
- **Potential energy** represents the energy stored in an object that has the *potential* to become another form of energy (usually kinetic)
    - **Kinetic energy** is the energy associated with an object's motion
        - Mechanical motion
        - Electron flow



ELECTRIC POTENTIAL

- **Electric potential energy** is the amount of work needed to move a charged particle between two points - measured in **Joules (J)**
- Charged particles move from **high to low** electric potential



Voltage

- **Voltage** is the **electric potential difference**
  - Current can flow where **voltage** exists
  - Measured in **Volts, V**
- **Batteries** are a voltage source
  - Conventional current will flow from the **positive (+)** to **negative (-)** terminal when connected.
    - The **positive (+)** terminal has a *higher* potential than the **negative (-)**



Voltage

- **Measuring voltage requires a reference point** because it is the difference in volts between two points

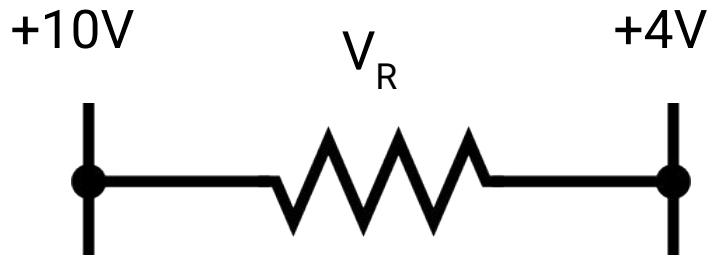
called “**Ground**”/”**GND**”

- Using the **negative (-)** terminal of the battery as a reference point:
  - The **positive (+)** terminal has a **voltage** of **+9V**
  - The **negative (-)** terminal has a **voltage** of **+0V**
    - **Why zero?** There is no potential difference at the same location as the reference point

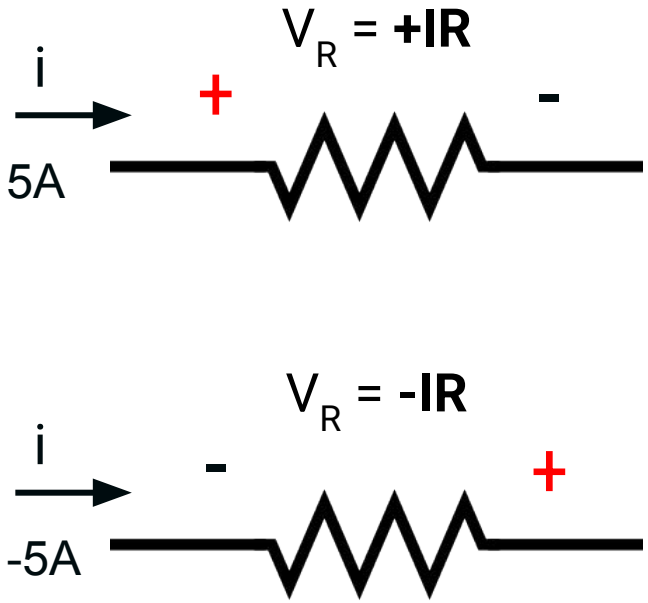


Voltage Drop

- **Voltage Drop** is the **difference of electric potential** (in volts) **between two positions,** following the path of current flowing in a circuit
  - An LED experiences a voltage drop because it expends energy to emit light
  - A resistor experiences a voltage drop because electrons lose potential energy as they go through it
- We can calculate the voltage drop between two nodes (across a circuit component)

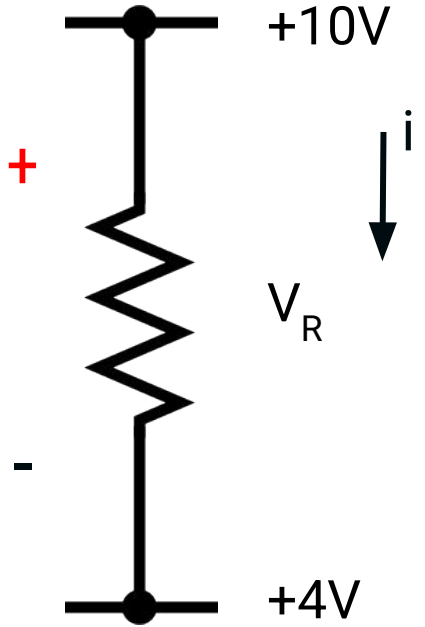


Voltage Drop Sign Convention (Cont'd)



- The voltage drop is **positive** if current flows from **+** to **-** terminals
  - Conventional current flows from **+** to **-** terminals (even though in reality it is the opposite)
- The voltage drop is **negative** if current flows from **-** to **+** terminals

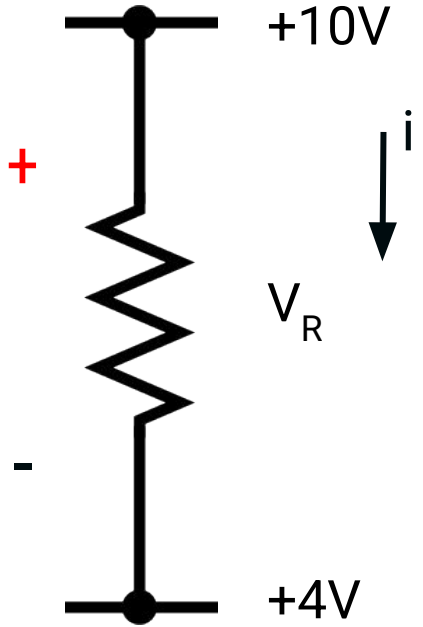
Calculating Voltage Drop



Find voltage drop $\mathbf{V_R}$

1. Calculate $\mathbf{V_R}$ as the **difference in volts** between terminals, following the direction of current

Calculating Voltage Drop (Cont'd)



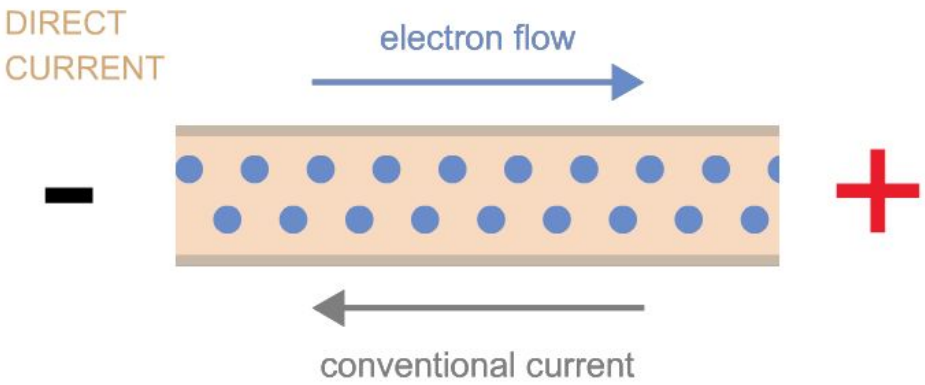
Find voltage drop $\mathbf{V_R}$

1. Calculate $\mathbf{V_R}$ as the **difference in volts** between terminals, following the direction of current

$$V_R = 10V - 4V = \boxed{6V}$$

Current

- **Current** is the flow of electrons
  - Current has a magnitude and direction, measured in **amperes** (**A**)
  - **Conventional current** flows in the direction *opposite* to the electron flow



How to Limit Current

We use resistors:

- A **resistor** is a component that reduces the flow of current
- **Resistance**, measured in **ohms (Ω)**, is the ability to *resist* current



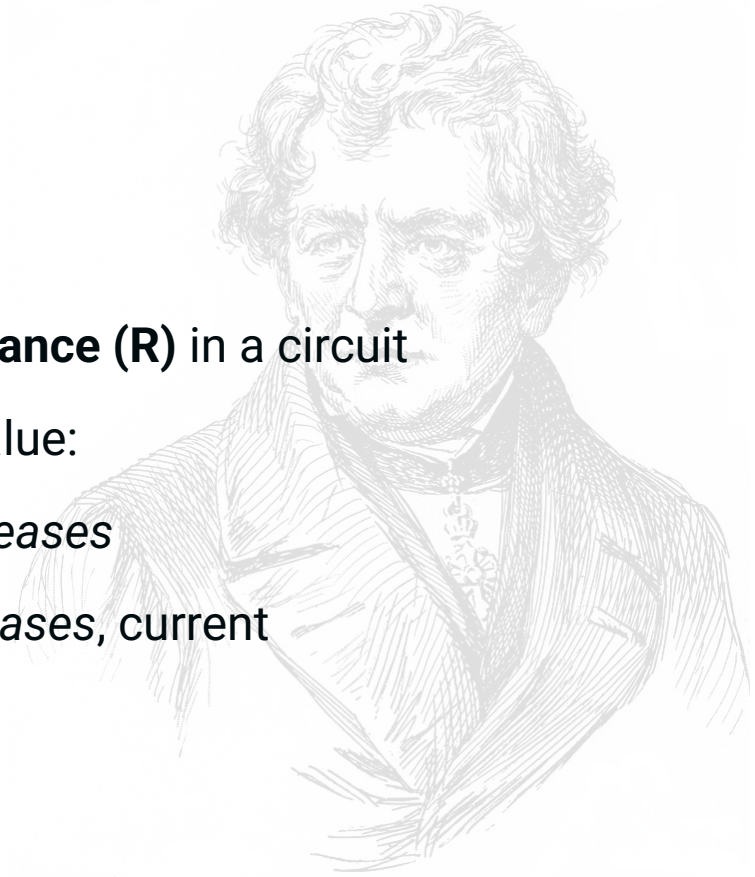
Why do we need Resistors?

- We need resistors to reduce/control the flow of current because components are usually designed with a specific current in mind
- If that current threshold is exceeded, then it could damage the component



Ohm's Law

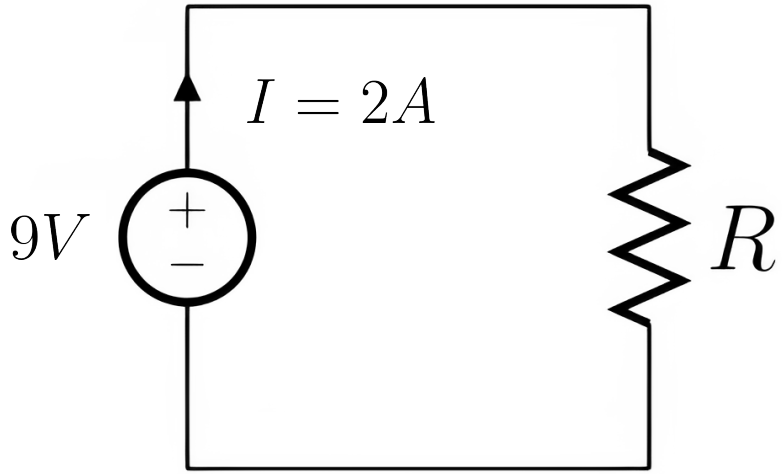
$$V = IR$$

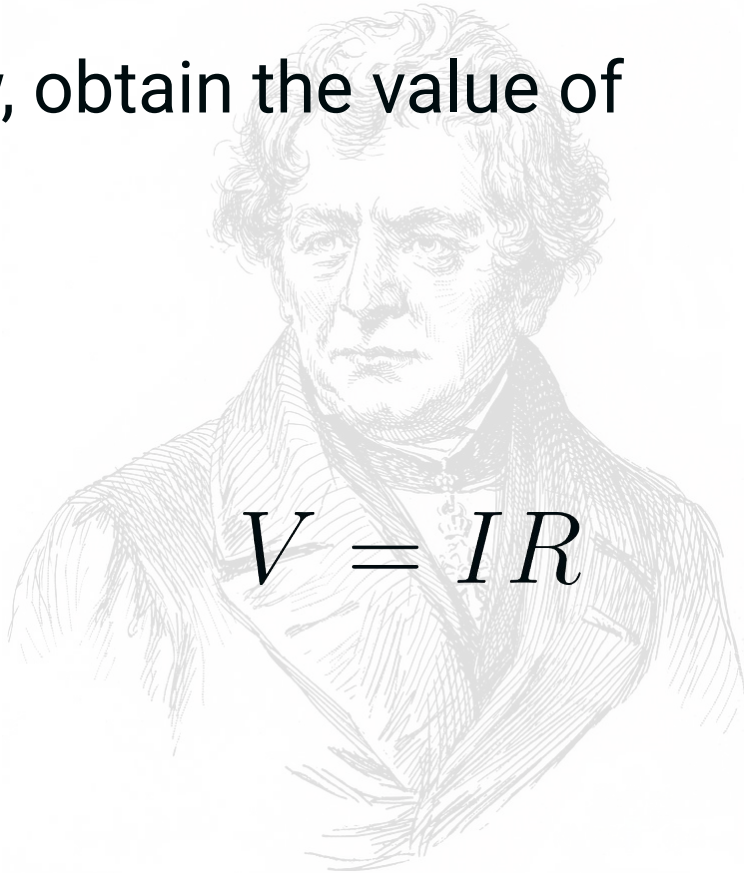


- Relates the **voltage (V)**, **current (I)**, and **resistance (R)** in a circuit
- Suppose we hold the voltage at a constant value:
  - As the resistance *increases*, current *decreases*
  - In the opposite case, as resistance *decreases*, current *increases*

Applying Ohm's Law

Given the circuit schematic below, obtain the value of resistor (**R**).

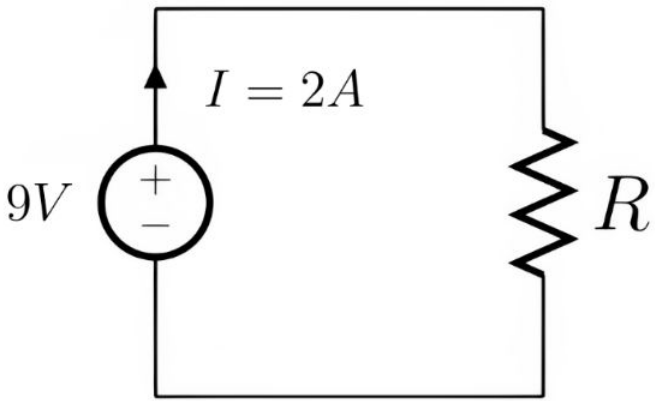




$$V = IR$$

Given the circuit schematic below, obtain the value of resistor (R).

$$V = IR$$



1.8

2.3

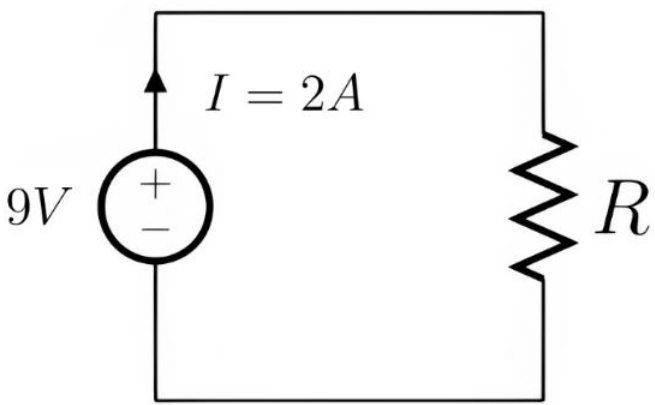
4.5

5.2

None of the above

Given the circuit schematic below, obtain the value of resistor (R).

$$V = IR$$



1.8

2.3

4.5

5.2

None of the above

SECTION II

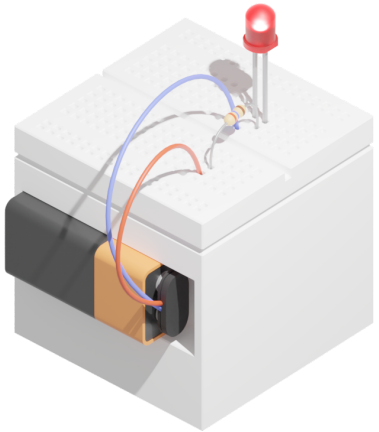
Project 1:
LED There Be Light

Project 1 Overview

- Build an **LED circuit** with a switch and solder it to a perfboard.

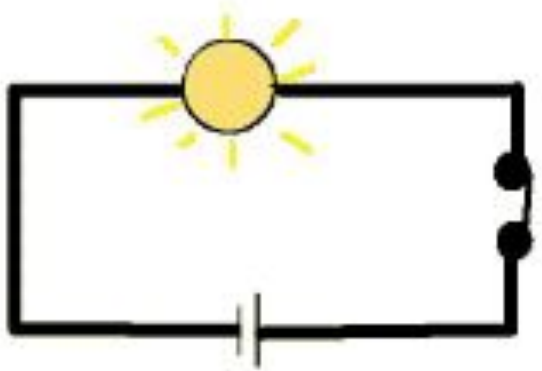
You will learn:

- Ohm’s Law
  - (Voltage, Current, Resistance)
- Breadboarding
- Soldering



Introduction to Circuits

- A **circuit** is a **closed loop** path where electrons can flow
  - The flow of electrons is commonly known as **current**.
- An **open circuit** **contains a discontinuity** that disrupts current
  - Ex) An open switch on a circuit prevents current from flowing
- A **closed circuit** has a **continuous** path for current to flow through



Circuit Components

- All circuits require the following three components:
  - A **voltage source** to provide electrical energy and generate current
    - Ex) DC power supply, batteries
  - A **conductive path** for current to flow
    - Ex) wire, copper, circuit board traces
  - A **load** to expend the electrical energy
    - Ex) light bulbs, sound speakers, motors, **occasionally you**

- For Project 1 (In your OPS Kit):
  - **Voltage source:** 9V battery



  - **Conductive path:** Jumper Wires

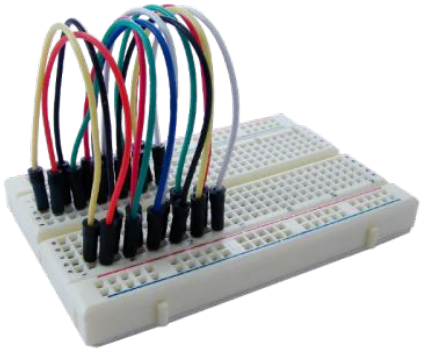


  - **Load:** LEDs, motors, resistors



How Current Flows Through a Circuit

- Current cannot flow without a path for the electrons to move
- A **conductor** is a material in which electrons can move freely
    - If a **voltage** exists across a conductor, current will flow
    - The opposite, **insulators**, restrict electron flow
- We connect circuits with jumper wires, which are conductive paths.





SECTION III

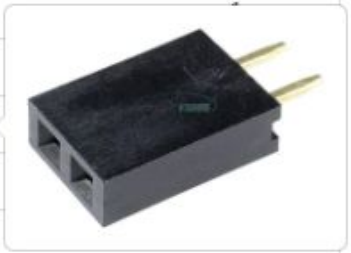
Project 1 Notes

Project 1 Components

- Battery, 9V
- Breadboard
- Header, 2.54mm, Female, 1x2
- Jumper Wires
- LED, 3.2V
- Perfboard
- Resistor
- Slide Switch
- Snap Connector, 9V

Parts

Part Name	Qty
Jumper Wire	?
Breadboard	1
Battery, 9V	
9V Snap Connector	
Header, 2.54mm, Female, 1x2	
LED, 2V	
Slide Switch	
Resistor, ?Ω	1
Perfboard	1

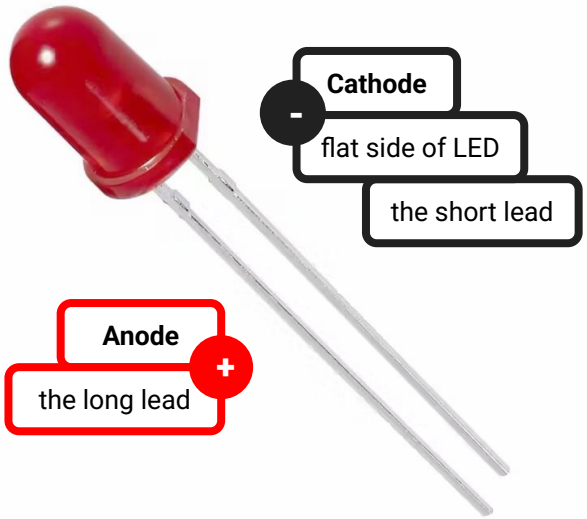


Polarized Components

- A **polarized component** has a positive and negative end, and can only be **connected** to the circuit **in one direction**
  - Ex) Batteries, LEDs, Electrolytic Capacitors
  - These components have a **positive** (**anode**) and **negative (cathode)** terminal
    - Remember that current flows from **positive** to **negative**
    - Terminals may be **distinguished by lead length**, **labels**, or **notches** on the component
    - Sketches of circuits will also indicate the components' polarities

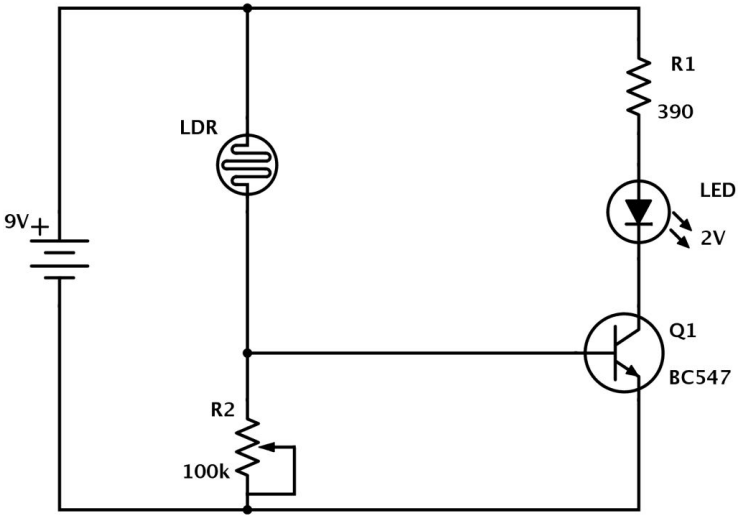
LEDs

- The **Light Emitting Diode (LED)** is a component that emits light (big surprise)
  - As a **diode**, the LED is **polarized** and has a constant **forward voltage** between its anode and cathode terminals (normal operating conditions)
  - Refer to the LED's **datasheet** to find its **operating conditions**



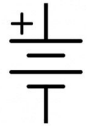
Schematics

- A **schematic** diagram defines the **connections between components**
- Schematics also **summarize the components' values**
- Each component has a **unique symbol** associated with it.

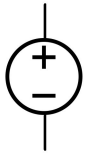


Schematic Symbols

Battery



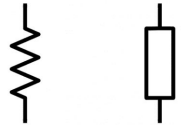
Voltage Source



Ground



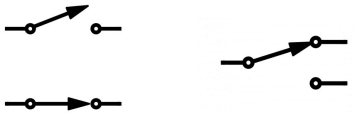
Resistor



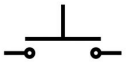
Potentiometer



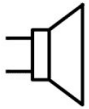
Switches



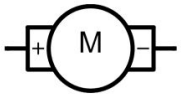
Pushbutton



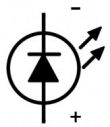
Speaker



DC Motor

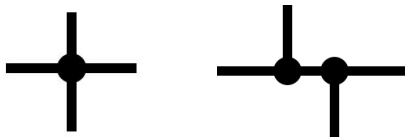


LED

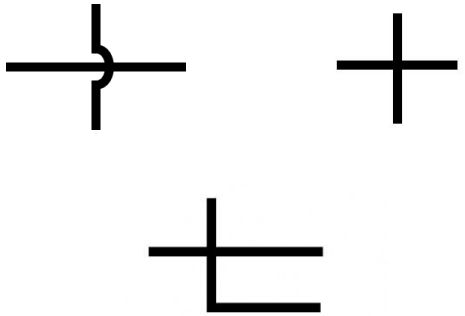


Schematic Connections

Connected Wires

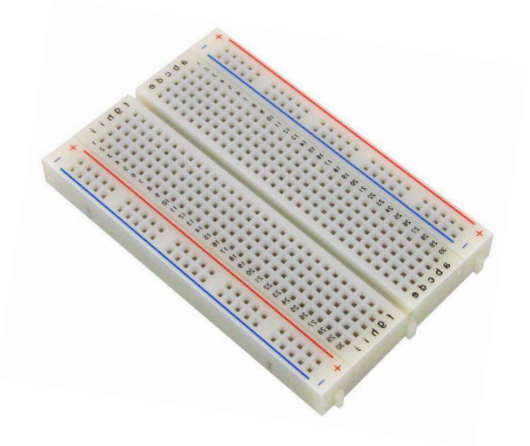


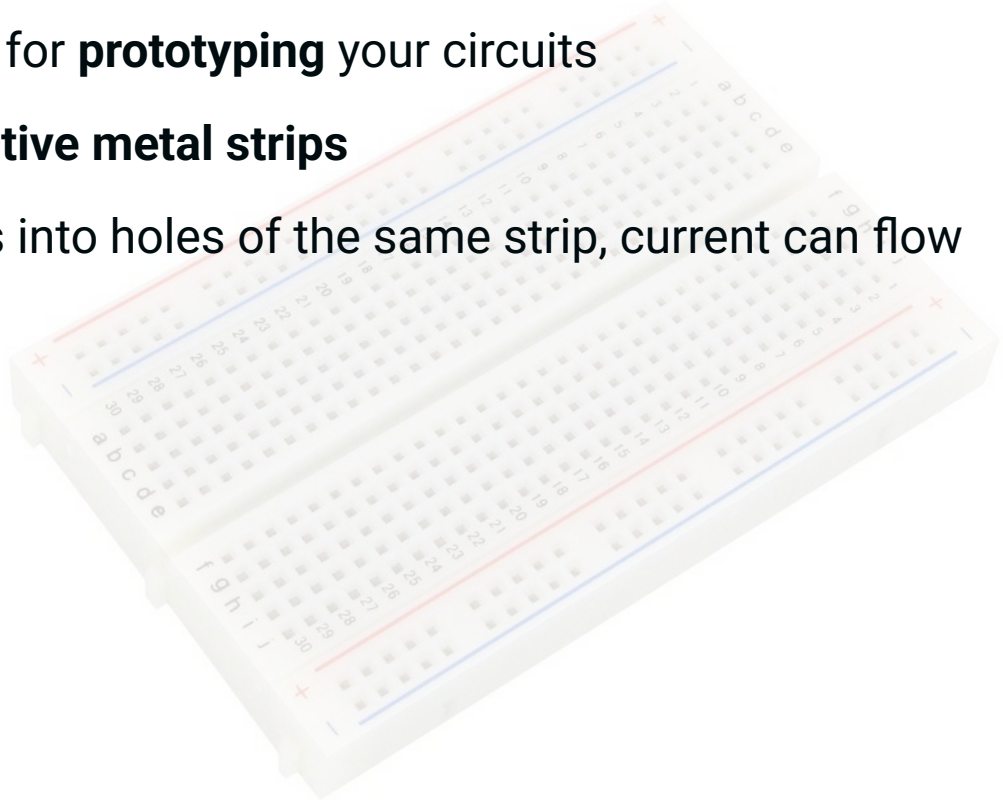
Unconnected Wires



Breadboards

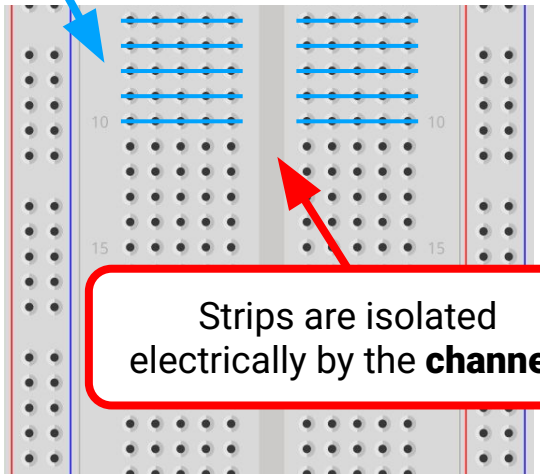
- A **breadboard** is a reusable board for **prototyping** your circuits
- Inside the breadboard are **conductive metal strips**
    - When you insert components into holes of the same strip, current can flow between those components





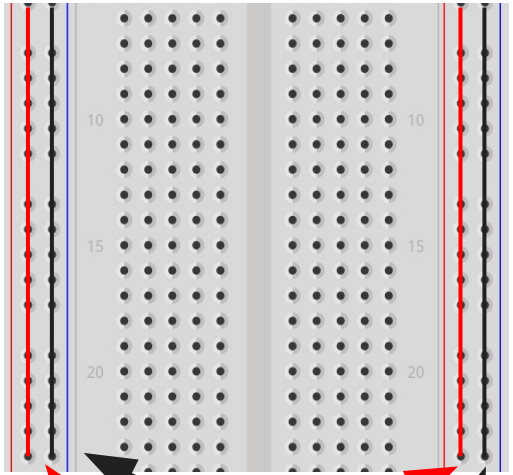
Breadboards

Horizontal holes are electrically connected in a **terminal strip**



Strips are isolated electrically by the **channel**

Vertical strips (**bus strips**) are connected down the entire board

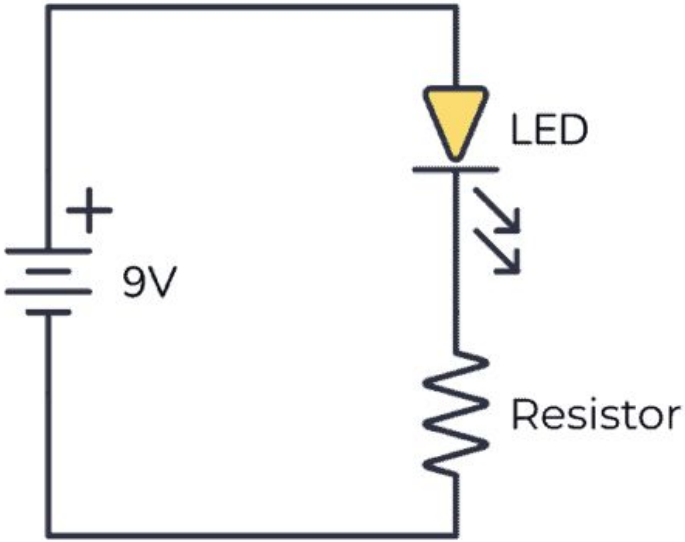


Power Rails **+**

Ground Rails **-**

Breadboard Exercise

- Let's set up a simple circuit on a breadboard with a 9V battery, LED, jumper wires, and a 390Ω resistor
- We have three components in this circuit - don't forget to place their terminals on **different** breadboard strips!
- Don't forget that the **longer** lead (anode) on the LED is **positive**, and the **shorter** lead (cathode) is **negative**
    - Remember that current flows from positive to negative - **don't flip the placement of your LED (it will fry)!**

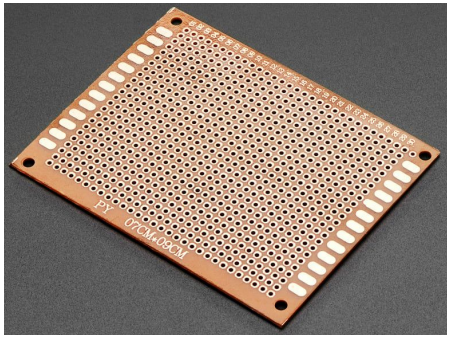


Soldering

- Preliminary circuit designs are built on breadboards
- Final circuit designs are often soldered onto **printed circuit boards (PCBs)** or **perfboards**
- **Soldering** is a process that joins circuit components together with a filler metal



Printed Circuit Board



Perfboard

Soldering Workshops

- If you would like to solder next week (Week 4), please sign up via **Google Sheets**
- OPS Workshop 1: Soldering
- Place your name in an empty slot! →

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